MASTER DOCUMENT TREE

Business / Personal Jet Host Variant – Full Vehicle Package

250 Engineering Documents

Ultra-Safe Central Capsule as Central Survival Core (5.80 m × 3.80 m | 22 t | 7 Occupants)

Aligned to Document 00 Version 3.0 — Comprehensive Engineering Baseline

Document Version	Date	Status	Total Documents
1.0	September 2026	Controlled Tree aligned to Document 00 Version 3.0	**250**

This Master Document Tree defines the complete set of 250 engineering documents required to design a business or personal jet that uses the Ultra-Safe Central Capsule (Large Configuration) as its central survival core. The capsule remains the pure survival module defined in Document 00 Version 1.6 (no under-surface rotors). The jet airframe, wing, propulsion, flight controls, landing gear, systems and separation interfaces are designed around the capsule. In-flight separation is performed by multi-path release followed by parachute recovery of the capsule. The package covers the complete vehicle from requirements through architecture, structures, aerodynamics, propulsion, avionics, safety, testing and certification considerations at the same depth as the Helicopter Host Package.

1. Document Package Overview by Domain

Group	Domain	Doc Range	Count
J1	Program Control, Requirements & Safety Case	001 – 030	30
J2	Vehicle Architecture & Systems Engineering	031 – 060	30
J3	Airframe, Wing & Capsule Integration Structures	061 – 100	40
J4	Aerodynamics, Propulsion & Flight Performance	101 – 140	40
J5	Separation, Safety Layers & Crashworthiness	141 – 175	35
J6	Avionics, Electrical, Flight Controls & Systems	176 – 205	30
J7	Landing Gear, Hydraulics & Mechanical Systems	206 – 225	20
J8	Human Factors, Cabin & Crew Systems	226 – 240	15
J9	Verification, Test, Certification & Support	241 – 250	10
	TOTAL	**001 – 250**	**250**

2. Complete Document Tree (001 – 250) with Document Types

All 250 documents are listed with number, full title and Document Type. The capsule remains the pure survival module (Document 00 Version 1.6). The jet is designed around it as the central survival core at the same documentation depth as the Helicopter Host Package.

No.	Document Title	Document Type
	GROUP J1 – Program Control, Requirements & Safety Case (001 – 030)	
1	Jet Host Master Document Index & Tree (this document)	Management
2	Jet Project Charter & Vision Statement	Management
3	Scope Definition – Business/Personal Jet with Capsule Core	Requirements
4	Stakeholder Needs & Operational Requirements	Requirements
5	Concept of Operations (ConOps) – Jet + Capsule	Operations
6	Design Reference Missions & Mission Profiles	Requirements
7	System Requirements Specification (SRS) – Top Level	Requirements
8	Functional Requirements Specification – Vehicle	Requirements
9	Performance Requirements Specification	Requirements
10	Safety Requirements Specification – Jet Host	Safety
11	Environmental Requirements Specification	Requirements
12	Human Factors Requirements – Flight Crew & Passengers	Requirements
13	Reliability, Availability, Maintainability Requirements	Requirements
14	Interface Requirements – Capsule to Jet	Requirements
15	Derived Requirements & Design Constraints	Requirements
16	Requirements Traceability Matrix – Jet Package	Management
17	Preliminary Hazard Analysis (PHA)	Safety
18	Functional Hazard Assessment (FHA)	Safety
19	System Safety Assessment Plan	Safety
20	Common Cause Analysis (CCA)	Safety
21	Aircraft-Level FMECA	Safety
22	Fault Tree Analysis – Critical Functions	Safety
23	Safety Case Part 1 – Argument Structure	Safety
24	Safety Case Part 2 – Evidence Framework	Safety
25	Airworthiness Certification Basis Mapping	Certification
26	Means of Compliance Matrix	Certification
27	Particular Risk Analysis (Impact, Fire, Cabin Depressurisation)	Safety
28	Configuration Management Plan	Management
29	Risk Management Plan & Initial Risk Register	Management
30	Program Schedule, Milestones & Resource Outline	Management
	GROUP J2 – Vehicle Architecture & Systems Engineering (031 – 060)	
31	Jet System Architecture Description	Architecture
32	Functional Architecture – Vehicle Level	Architecture
33	Physical Architecture & Product Breakdown Structure	Architecture
34	Logical Architecture & Data Flow	Architecture
35	Mode & State Definition Document	Architecture
36	Operational Mode Transition Matrix	Architecture
37	Capsule-to-Jet Interface Control Document (Master)	ICD
38	Internal Vehicle Interface Control Document	ICD
39	Electrical Power Architecture – Jet	Architecture
40	Hydraulic System Architecture	Architecture

No.	Document Title	Document Type
41	Data Architecture & Avionics Network	Architecture
42	Redundancy & Fail-Operational Architecture	Architecture
43	Six-Layer Safety Architecture Integration with Jet	Architecture
44	Separation Sequence Logic & Timing (In-Flight)	Design
45	Flight Control Architecture	Architecture
46	Autonomy & Flight Management Architecture	Architecture
47	Health Monitoring & Prognostics Architecture	Architecture
48	Vehicle Mass Properties Control Plan	Management
49	Centre of Gravity & Inertia Management – Complete Vehicle	Analysis
50	Thermal Architecture – Jet	Architecture
51	Electromagnetic Compatibility Architecture	Architecture
52	Cyber-Physical Security Architecture	Architecture
53	Model-Based Systems Engineering Model Description	Architecture
54	Trade Study Log – Configuration Decisions	Analysis
55	System Performance Budgets (Mass, Power, Volume, Timing)	Analysis
56	System Integration Strategy	Management
57	End-to-End Mission Timeline – Jet Operations	Operations
58	Open Issues & Technical Debt Register	Management
59	Design Review Process & Checklist Library	Management
60	Systems Engineering Management Plan – Jet	Management
	GROUP J3 – Airframe, Wing & Capsule Integration Structures (061 – 100)	
61	Airframe Structural Design Philosophy & Criteria	Requirements
62	Primary Airframe Structure Design Description	Design
63	Capsule Bay Structure Design	Design
64	Capsule Attachment Hard-Points Design (Shock-Absorbing)	Design
65	Load Path Analysis – Capsule to Airframe	Analysis
66	Fuselage Structure Design	Design
67	Wing Structure Design Description	Design
68	Wing-Fuselage Carry-Through Structure Design	Design
69	Empennage Structure Design	Design
70	Landing Gear Attachment Structure Design	Design
71	Engine Mount Structure Design	Design
72	Door, Hatch & Access Panel Structural Design	Design
73	Material Selection Report – Airframe & Wing	Materials
74	Material Allowables & Design Values	Materials
75	Joint Design & Fastener Specification	Specification
76	Corrosion Protection & Coatings Specification	Specification
77	Airframe & Wing Mass Breakdown & Optimization	Analysis
78	Finite Element Model Description – Global Airframe	Analysis
79	Finite Element Model – Capsule Integration Region	Analysis
80	Static Load Analysis Report	Analysis
81	Dynamic & Modal Analysis Report	Analysis

No.	Document Title	Document Type
82	Crash & Impact Analysis – Airframe + Capsule	Analysis
83	Fatigue & Damage Tolerance Analysis	Analysis
84	Aeroelastic Analysis (Flutter, Divergence)	Analysis
85	Buckling & Stability Analysis	Analysis
86	Structural Safety Factor Justification	Analysis
87	Damage Tolerance & Fail-Safe Assessment	Safety
88	Residual Strength after Capsule Separation	Analysis
89	Structural Test Requirements Specification	Test
90	Structural Test Plan – Static	Test
91	Structural Test Plan – Fatigue & Damage Tolerance	Test
92	Capsule-to-Airframe Interface Strength Test Plan	Test
93	Critical Structural Items List – Jet	Management
94	Structural Design Review Checklist	Review
95	Weight & Balance Report – Complete Jet	Analysis
96	Inertia & Mass Distribution Report	Analysis
97	Structural Interface Control Document – Capsule Bay	ICD
98	Manufacturing Considerations – Airframe & Wing	Manufacturing
99	Assembly Sequence – Capsule into Airframe	Manufacturing
100	Airframe Family Document Index	Management
	GROUP J4 – Aerodynamics, Propulsion & Flight Performance (101 – 140)	
101	Aerodynamic Design Requirements	Requirements
102	Aerodynamic Configuration Description	Architecture
103	Wing Aerodynamic Design	Design
104	High-Lift System Design	Design
105	Empennage Aerodynamic Design	Design
106	Computational Fluid Dynamics Analysis Plan	Analysis
107	Aerodynamic Performance Analysis	Analysis
108	Stability & Control Derivatives Analysis	Analysis
109	Propulsion System Requirements	Requirements
110	Engine Selection & Installation Design	Design
111	Nacelle & Inlet Design	Design
112	Thrust Reverser / Exhaust System Design	Design
113	Fuel System Design Description	Design
114	Propulsion System Performance Analysis	Analysis
115	Propulsion System FMECA	Safety
116	Flight Performance Requirements	Requirements
117	Performance Analysis – Take-off, Climb, Cruise, Landing	Analysis
118	Range & Payload Analysis	Analysis
119	Flight Envelope Definition	Requirements
120	Handling Qualities Requirements	Requirements
121	Stability & Control Analysis	Analysis
122	One-Engine-Inoperative Performance	Analysis

No.	Document Title	Document Type
123	Noise Requirements & Assessment	Requirements
124	Climb Gradient & Obstacle Clearance Analysis	Analysis
125	Propulsion & Aerodynamics Interface Control Document	ICD
126	Aerodynamic Test Plan Outline (Wind Tunnel / CFD)	Test
127	Propulsion System Test Plan Outline	Test
128	Performance Flight Test Requirements	Test
129	Aerodynamics Critical Design Review Checklist	Review
130	Propulsion Critical Design Review Checklist	Review
131	Performance Validation Plan	Test
132	Engine Bird Strike & Ingestion Considerations	Safety
133	Fuel System Safety Assessment	Safety
134	Thrust Asymmetry & Control Analysis	Analysis
135	Aerodynamic Family Risk Assessment	Safety
136	Propulsion Family Risk Assessment	Safety
137	Engine Control System Interface Requirements	Requirements
138	Nacelle Structural & Thermal Design	Design
139	Performance Sensitivity Study	Analysis
140	Aerodynamics & Propulsion Family Document Index	Management
	GROUP J5 – Separation, Safety Layers & Crashworthiness (141 – 175)	
141	In-Flight Separation System Requirements	Requirements
142	Separation Architecture – Jet Host	Architecture
143	Pyrotechnic Separation Path Design	Design
144	Hydraulic / Mechanical Separation Path Design	Design
145	Pure Mechanical Manual Separation Design	Design
146	Separation Sequence Timing & Interlocks – Flight	Design
147	Clean-Break Disconnect Design – All Interfaces	Design
148	Separation Dynamics Analysis – In-Flight Release	Analysis
149	Re-contact & Structural Interference Risk Assessment	Safety
150	Separation System FMECA – Jet	Safety
151	Crew-Commanded vs Automatic Separation Criteria	Requirements
152	Integration of Capsule Six Safety Layers with Jet	Architecture
153	Layer 1–4 Activation Logic in Jet Context	Design
154	Post-Separation Capsule Trajectory & Clearance Analysis	Analysis
155	Crashworthiness Requirements – Complete Vehicle	Requirements
156	Crash Load Cases – Jet + Capsule	Safety
157	Energy Absorption Strategy – Airframe & Landing Gear	Design
158	Occupant Protection under Jet Crash Loads	Analysis
159	Fire Protection System – Jet Airframe	Design
160	Fire Detection & Suppression Logic – Vehicle Level	Design
161	Cabin Depressurisation & Emergency Descent Considerations	Safety
162	Ditching Performance & Capsule Release after Ditching	Analysis
163	Safety Assessment – Combined Jet + Capsule	Safety

No.	Document Title	Document Type
164	Particular Risks – Engine Fire, Fuel Leak, Rapid Depressurisation	Safety
165	Separation System Interface Control Document	ICD
166	Crashworthiness Test Plan Outline	Test
167	Separation System Test Plan – Ground & Flight	Test
168	Safety Critical Design Review Checklist	Review
169	Airworthiness Safety Case Contribution – Separation	Certification
170	Emergency Procedures – Capsule Separation in Flight	Operations
171	Post-Separation Recovery Coordination Procedures	Operations
172	Safety Family Risk Register – Jet Host	Safety
173	Independence Demonstration Plan – Separation Paths	Safety
174	Certification Considerations – Separation System	Certification
175	Separation & Safety Family Document Index	Management
	GROUP J6 – Avionics, Electrical, Flight Controls & Systems (176 – 205)	
176	Avionics Architecture Description – Jet	Architecture
177	Electrical Power System Design – Vehicle Level	Design
178	Emergency Power & Battery Bus Integration with Capsule	Design
179	Flight Control System Architecture	Architecture
180	Flight Control Computer Specification	Specification
181	Primary Flight Control Laws & Modes	Design
182	Automatic Flight Control System Requirements	Requirements
183	Sensor Suite Requirements – Jet	Requirements
184	Navigation System Requirements	Requirements
185	Communication System Requirements	Requirements
186	Display & Crew Alerting System Requirements	Requirements
187	Human-Machine Interface Design – Cockpit	Design
188	Health Monitoring System Design	Design
189	Mission / Cabin System Architecture	Architecture
190	Data Bus & Network Architecture	Architecture
191	Software Architecture & Partitioning – Vehicle	Software
192	Software Development Assurance Levels	Software
193	Software Requirements – Safety Critical Functions	Software
194	Electromagnetic Compatibility Requirements – Vehicle	Requirements
195	Lightning Protection Design	Design
196	Cybersecurity Requirements – Flight Critical Systems	Security
197	Avionics Environmental Qualification Requirements	Requirements
198	Avionics Interface Control Document	ICD
199	Avionics Mass & Power Budget	Analysis
200	Avionics Test & Verification Plan Outline	Test
201	Flight Control System Test Plan Outline	Test
202	Avionics Family Risk Assessment	Safety
203	Avionics Critical Design Review Checklist	Review
204	Cockpit Design Review Checklist	Review

No.	Document Title	Document Type
205	Avionics & Flight Control Family Document Index	Management
	GROUP J7 – Landing Gear, Hydraulics & Mechanical Systems (206 – 225)	
206	Landing Gear Requirements Specification	Requirements
207	Landing Gear Architecture & Configuration	Architecture
208	Main Landing Gear Design Description	Design
209	Nose Landing Gear Design Description	Design
210	Landing Gear Shock Absorption Analysis	Analysis
211	Landing Gear Loads & Strength Analysis	Analysis
212	Landing Gear Retraction / Extension System Design	Design
213	Hydraulic System Design Description	Design
214	Hydraulic Power Generation & Distribution	Design
215	Flight Control Hydraulic Actuation Design	Design
216	Utility Hydraulic Systems Design	Design
217	Brake System Design	Design
218	Landing Gear & Hydraulics FMECA	Safety
219	Landing Gear Interface Control Document	ICD
220	Landing Gear Test Plan Outline	Test
221	Hydraulic System Test Requirements	Test
222	Landing Gear Critical Design Review Checklist	Review
223	Ground Handling & Towing Provisions	Design
224	Landing Gear Family Risk Assessment	Safety
225	Landing Gear & Hydraulics Family Document Index	Management
	GROUP J8 – Human Factors, Cabin & Crew Systems (226 – 240)	
226	Human Factors Engineering Requirements – Jet	Requirements
227	Cockpit Layout & Ergonomics Design	Design
228	Passenger Cabin Layout (around Capsule Core)	Design
229	Crew & Passenger Seat Design	Design
230	Restraint System Design	Design
231	Emergency Egress Design – Jet Airframe	Design
232	Egress Path Analysis after Capsule Separation	Analysis
233	Internal Lighting, Marking & Wayfinding	Design
234	Cabin Comfort, Noise & Environmental Control	Requirements
235	Crew Workload Assessment Methodology	Analysis
236	Human Factors Test & Evaluation Plan Outline	Test
237	Occupant Injury Criteria – Jet Crash	Safety
238	Crew Training & Emergency Procedure Outline	Operations
239	Human Factors Critical Design Review Checklist	Review
240	Human Factors Family Document Index	Management
	GROUP J9 – Verification, Test, Certification & Support (241 – 250)	
241	Jet Verification & Validation Master Plan	Test
242	Ground Test Plan Outline – Complete Vehicle	Test
243	Flight Test Plan Outline	Test

No.	Document Title	Document Type
244	Integrated Capsule Separation Flight Test Requirements	Test
245	Certification Program Plan – Jet Host	Certification
246	Type Certification Data Sheet Outline	Certification
247	Continued Airworthiness & Maintenance Concept	Support
248	Spare Parts & Support Equipment Requirements	Support
249	Jet Host Risk Register – Master	Management
250	Jet Host Package Document Index & Traceability	Management

This 250-document Jet Host Variant Package is controlled against Document 00 Version 1.6. The Ultra-Safe Central Capsule remains the pure survival module (no under-surface rotors). The jet airframe, wing, propulsion and all vehicle systems are designed around the capsule as the central survival core. In-flight separation is multi-path followed by parachute recovery of the capsule. Documentation depth matches the Helicopter Host Package. All numerical values, safety principles and interface rules shall remain consistent with the locked baseline of the capsule.

— End of Jet Host Variant Master Document Tree (250 Documents) —
Ultra-Safe Central Capsule | Document 00 Version 1.6 | Jet Host Package